

Indian Institute Of Information Technology Guwahati



Project Report 3

Group Number - 14

Ansik Singh Tomar (2401037)

Anuved Pratap Singh (2401041)

Anmol Kumar (2401031)

Anmol Kumar (2401032)

Section: CS21

Branch: B.Tech 2nd Year CSE

Course: DBMS Lab

Code: CS241

Project Delivery

We developed a web app using react for the frontend, and Spring boot for the backend, and mySQL for the database. We have an functional integrated Admin Panel where the Admin can view the stats and in a dedicated custom Query panel he can Query the database.

Individual Contribution

1. Ansik Singh Tomar (2401037)

- Architected the React frontend infrastructure and global state management.
- Integrated JWT-based authentication flow with silent refresh token logic, to the frontend
- Integrated the APIs developed by the backend team with the react components.
- Developed the dynamic Admin and Agent dashboards with real-time analytics visualization.

2. Anmol Kumar (2401032)

- Designed the complete backend system architecture and developed scalable REST APIs using Spring Boot following layered architecture
- Implemented secure authentication and authorization using JWT with role-based access control for User, Admin, Office, and Agent modules
- Applied Strategy Design Pattern to build flexible API architecture, enabling easier frontend integration and extensibility
- Developed analytics and revenue tracking system with optimized and complex database queries for efficient data processing and reporting.

1. Anmol Kumar (2401031)

- Designed intuitive and user-friendly interfaces with a focus on usability and accessibility
- Created wireframes, prototypes, and user flows to enhance user experience across the platform
- Collaborated with developers to ensure seamless implementation of design systems
- Conducted user research and usability testing to improve interaction and visual design.

4. Anuved Pratap Singh (2401041)

- Architected a robust relational database schema for properties, listings, agents, and transactions
- Engineered RESTful APIs for property, listing token, and image management with a focus on scalability
- Built and maintained data scraping workflows ensuring consistency, normalization, and reliability of data
- Enhanced query performance by optimizing complex SQL operations used in analytics and query tools

Opensource Libraries

- We used some libraries like, Shad.cn, Aceternity.ui, radix.ui, to make the react components interactive

AI usage - We used ai majorly for the debugging process, to solve the errors.